SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name	Dr. Rishabh Mittal		
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CourseCode	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/II	**Regulation**	R23
Date and Day of Assignment	**Week7 – Friday**	**Time(s)**	23CSBTB01 To 23CSBTB52
Duration	2 Hours	**Applicable to Batches**	All batches
Assignment Number: 14.5(Present assignment number)/**24**(Total number of assignments)			

	Question	***Expected Time to compl***

			ete
		Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation **Lab Objectives** • Design **functional, visually appealing** web applications using HTML, CSS, and JavaScript with AI assistance. • Apply **responsive, accessible, and interactive design principles**. • Create **practical UI components** for real-world web applications. • Use AI to optimize **layout, UX, and performance**.	

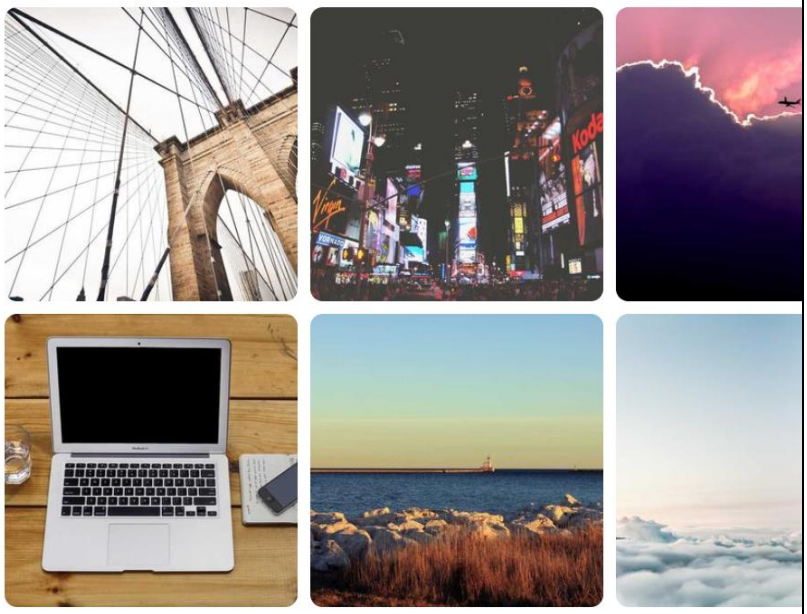
1		**Task 1 – Image Gallery Website**
		Create an image gallery web page.
		Requirements:
		• Display images in a grid layout.
		• Use AI to:
		  ○ Optimize layout using CSS Grid.
		  ○ Add hover zoom or overlay effects.
		  ○ Ensure mobile responsiveness.
		PROMPT
		Create a responsive image gallery using HTML and CSS Grid.
		Add hover zoom effects and make the layout mobile responsive.
		Use internal CSS.
		
		CODE
			Week7 - Friday

```
<!DOCTYPE html>
<html>
<head>
<title>Image Gallery</title>

<style>

body{
font-family: Arial;
text-align:center;
}

.gallery{
```

```
display:grid;
grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
gap:10px;
padding:20px;
}

.gallery img{
width:100%;
border-radius:10px;
transition:0.3s;
}

.gallery img:hover{
transform:scale(1.1);
}

</style>
</head>

<body>

<h2>Image Gallery</h2>

<div class="gallery">
<img src="https://picsum.photos/300?1">
<img src="https://picsum.photos/300?2">
<img src="https://picsum.photos/300?3">
<img src="https://picsum.photos/300?4">
<img src="https://picsum.photos/300?5">
<img src="https://picsum.photos/300?6">
</div>

</body>
</html>
```

OUTPUT

Image Gallery



EXPLANATION

- **CSS Grid** creates the gallery layout.
- `auto-fit` makes it responsive.
- `transform: scale(1.1)` creates hover zoom effect.

Task 2 – Profile Card Generator

Create a profile card web page.

Requirements:

- Display user image, name, role, and social media links.
- Use AI to:
  - Suggest modern UI colors and fonts.
  - Design the card using Flexbox.
  - Add hover animation and shadow effects

PROMPT

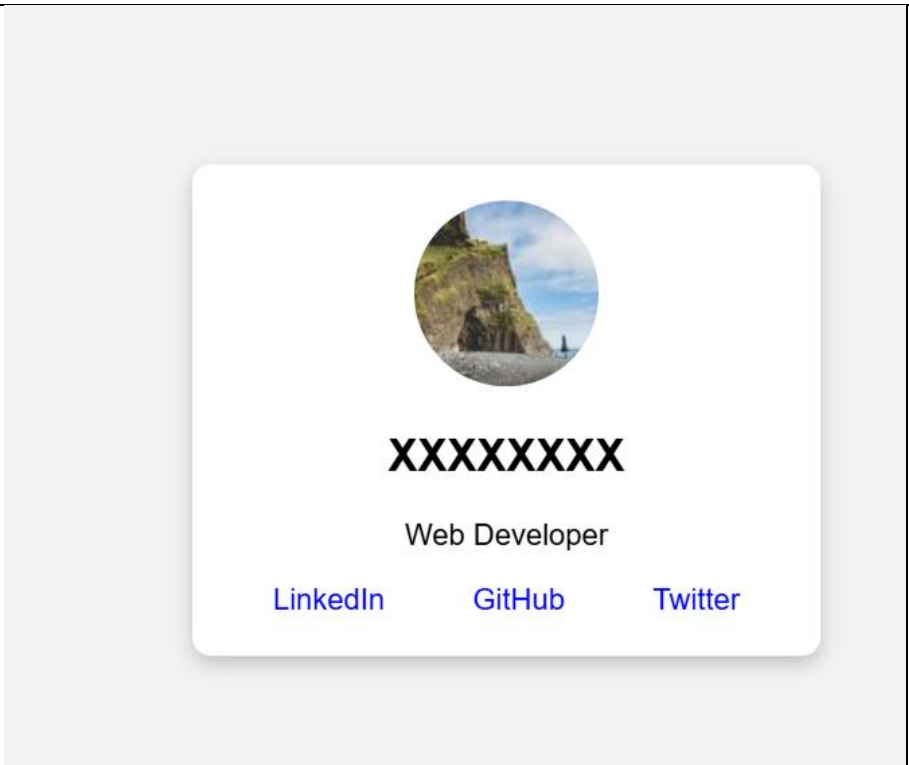
Create a modern profile card using HTML and Flexbox.
Include image, name, role and social links.
Add hover animation and internal CSS.

CODE

```html
<!DOCTYPE html>
<html>
<head>

<style>

body{
font-family:Arial;
background:#f2f2f2;
}

.card{
width:300px;
margin:100px auto;
padding:20px;
text-align:center;
border-radius:10px;
background:white;
box-shadow:0 4px 10px rgba(0,0,0,0.2);
transition:0.3s;
}

.card:hover{
transform:translateY(-10px);
}

.profile{
width:100px;
border-radius:50%;
}

.social{
display:flex;
justify-content:space-around;
margin-top:10px;
}

a{
text-decoration:none;
color:blue;
}
```

```html
</style>

</head>

<body>

<div class="card">

<img src="https://picsum.photos/100" class="profile">

<h2>XXXXXXXX</h2>
<p>Web Developer</p>

<div class="social">
<a href="#">LinkedIn</a>
<a href="#">GitHub</a>
<a href="#">Twitter</a>
</div>

</div>

</body>
</html>
```

OUTPUT



EXPLANATION

- *Flexbox aligns social links.*
- *box-shadow creates depth.*
- *Hover effect moves card upward.*

Task 3 -Develop a dynamic shopping cart system.
Requirements:

- Add, remove, and update product quantities.
- Display real-time total price.
- Use AI to:
  - Manage cart state using JavaScript objects.
  - Persist cart data in localStorage.

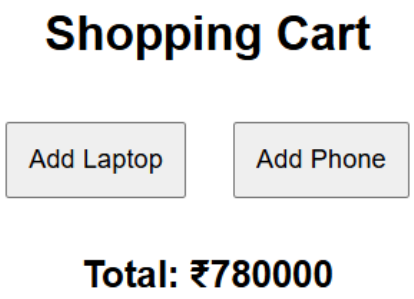
PROMPT

Build a simple shopping cart using HTML and JavaScript.Allow adding products and calculate total price dynamically.Use internal CSS.

CODE

```html
<!DOCTYPE html>
<html>
<head>

<style>
body{
font-family:Arial;
text-align:center;
}

button{
padding:10px;
margin:10px;
}
</style>

</head>

<body>

<h2>Shopping Cart</h2>

<button onclick="addItem(50000)">Add Laptop</button>
<button onclick="addItem(20000)">Add Phone</button>

<h3>Total: ₹<span id="total">0</span></h3>

<script>

let total=0;

function addItem(price){
total+=price;
document.getElementById("total").innerText=total;
}

</script>

</body>
</html>
```

OUTPUT



EXPLANATION

1. JavaScript function updates total price.
2. `innerText` changes displayed value dynamically.

Task 4 – Role-Based Access Control Application

Create a role-based web application.

Requirements:

- Admin, Editor, User roles.
- Conditional UI rendering.
- Use AI to:
  - Design access control logic.
  - Maintain secure state transitions.
  - Create scalable UI components.

PROMPT

Create a role based access system in HTML and JavaScript.
Include Admin, Editor and User roles.
Show UI based on selected role.

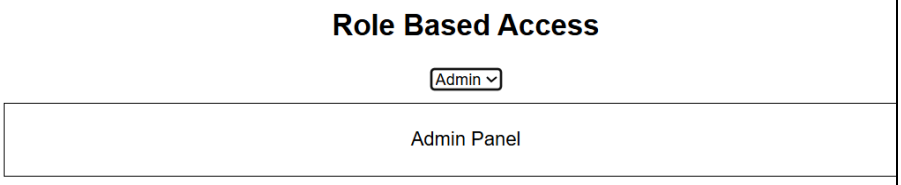
CODE

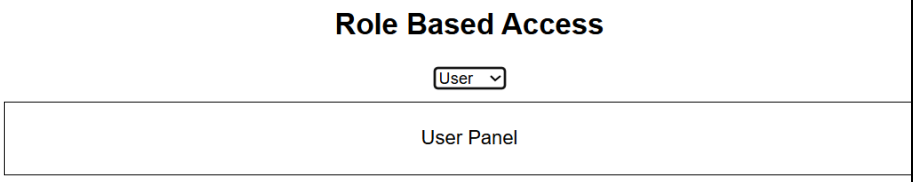
```
<!DOCTYPE html>
```

```html
<html>
<head>

<style>
body{
font-family:Arial;
text-align:center;
}

.panel{
display:none;
padding:20px;
margin:10px;
border:1px solid black;
}

</style>

</head>

<body>

<h2>Role Based Access</h2>

<select onchange="checkRole(this.value)">
<option value="admin">Admin</option>
<option value="editor">Editor</option>
<option value="user">User</option>
</select>

<div id="admin" class="panel">Admin Panel</div>
<div id="editor" class="panel">Editor Panel</div>
<div id="user" class="panel">User Panel</div>

<script>

function checkRole(role){

document.getElementById("admin").style.display="none";
document.getElementById("editor").style.display="none";
document.getElementById("user").style.display="none";

document.getElementById(role).style.display="block";
```

```
}

</script>

</body>
</html>
```

OUTPUT





EXPLANATION

- Role selection controls UI visibility.
- JavaScript dynamically shows correct panel.

Task 5 – Multi-Language Website

Design a multilingual website.

Requirements:

- Switch languages dynamically.
- Persist language preference.

- Use AI to:
  - Structure translation files.
  - Handle RTL/LTR layouts.
  - Improve accessibility.

PROMPT

Create a simple multilingual website using HTML and JavaScript.

Allow users to switch between English and Hindi.

Store language preference.

CODE

```
<!DOCTYPE html>
<html>
<head>

<style>
body{
font-family:Arial;
text-align:center;
}
</style>

</head>

<body>

<h2 id="title">Welcome</h2>

<button onclick="changeLang('en')">English</button>
<button onclick="changeLang('hi')">Hindi</button>

<script>

const translations={
en:{title:"Welcome"},
hi:{title:"स्वागत है"}
};
```

```
function changeLang(lang){

document.getElementById("title").innerText=
translations[lang].title;

localStorage.setItem("lang",lang);

}

</script>

</body>
</html>
```

OUTPUT

EXPLANATION

- Translation stored in **JavaScript object**.
- `localStorage` saves user language preference.

Task 6 – Online Examination System

Create an online exam interface.

Requirements:

- Timer, questions, and submission.
- Prevent multiple submissions.
- Use AI to:

- Manage exam state.
- Improve exam UX.
- Ensure accessibility.

PROMPT

Create an online exam interface using HTML and JavaScript.

Include timer, question display and submission button.

Prevent multiple submissions.

CODE

```html
<!DOCTYPE html>
<html>
<head>

<style>
body{
font-family:Arial;
text-align:center;
}
</style>

</head>

<body>

<h2>Online Exam</h2>

<p id="timer">Time:60</p>

<p>What is 2 + 2 ?</p>

<input type="radio" name="q1">3
<input type="radio" name="q1">4

<br><br>

<button onclick="submitExam()">Submit</button>

<script>
```

```
let time=60;
let submitted=false;

setInterval(function(){

time--;
document.getElementById("timer").innerText="Time:"+time;

if(time==0){
submitExam();
}

},1000);

function submitExam(){

if(submitted){
alert("Already submitted");
return;
}

submitted=true;
alert("Exam Submitted");

}

</script>

</body>
</html>
```

OUTPUT

Online Exam

Time:39

What is 2 + 2 ?

○ 3 ◉ 4

Submit ```

E***XPLANATION***

- Timer decreases every second.
- Exam auto-submits when time ends.
- Multiple submissions prevented.